

Introduction to Machine Learning

What Is Machine Learning?

Machine learning is a branch of artificial intelligence in which algorithms learn patterns from data rather than following explicitly programmed rules. The term was popularized by Arthur Samuel in 1959.

Types of Learning

Supervised learning uses labeled data to train models for tasks like classification and regression. Examples include spam detection and house price prediction.

Unsupervised learning finds hidden structure in unlabeled data, for instance clustering customers into segments using k-means. Reinforcement learning trains an agent to maximize cumulative reward through trial and error, as in game-playing systems.

Algorithm	Type	Typical Use
Linear Regression	Supervised	Price prediction
K-Means	Unsupervised	Customer segmentation
Random Forest	Supervised	Fraud detection
Q-Learning	Reinforcement	Game agents

Common Algorithms

Linear regression models the relationship between variables with a straight line. Decision trees split data using a sequence of questions, and random forests combine many trees to reduce overfitting. Neural networks stack layers of interconnected nodes and power modern deep learning systems.

Overfitting and Evaluation

Overfitting occurs when a model memorizes training data and fails to generalize. Common remedies include cross-validation, regularization, dropout, and collecting more data. Typical evaluation metrics are accuracy, precision, recall, and F1 score. A standard split is 80 percent training and 20 percent testing.